

ClaimsAI: AI-Powered Vehicle Damage Assessment

Product Requirements Document | Hossein Yousefi | June 2026

1. Vision & Goals

ClaimsAI replaces manual vehicle damage assessment with an AI-first workflow that reduces processing time by 90% while keeping humans in the loop. Designed for **regulated and government-administered insurance markets**, including Gulf states where QCB (Qatar) and SAMA (Saudi Arabia) mandate strict claims oversight and digital transformation aligns with Qatar National Vision 2030, as well as state-run programs like SAAQ (Quebec) and ICBC (British Columbia). Arabic-first multilingual support and in-country data residency are baseline requirements.

Goals: (1) Assessment time from ~45 min to <5 min. (2) AI estimates within 10% of final cost for 80%+ of claims. (3) Agent throughput from ~12 to ~50 claims/day. (4) Meet government audit, data residency, and RTL/Arabic requirements.

Competitive Context

Tractable leads in AI auto damage (NA/Europe); **CCC Intelligent Solutions** owns the dominant US repair database; **Mitchell** (Enlyte) provides integrated estimating. None have Gulf/MENA presence or offer **Arabic-first interfaces, sovereign cloud deployment, or Najm-style claims integration**. Our differentiation: a **data flywheel powered by Scale AI's labeling platform**, calibrated to each market's vehicle mix (Gulf skews luxury/SUV), regional labor rates, and regulatory requirements.

2. User Stories

Role	User Story	Acceptance Criteria
Claims Agent	Upload damage photos and receive instant AI assessment	Assessment within 30s. Arabic-first UI with English/French toggle.
Claims Agent	Review AI confidence to know when to trust or override	Confidence breakdown per area. Low-confidence flags visible.
Claims Agent	Annotate damage regions (confirm/reject/add) to improve model	Annotations enter training pipeline. Override rate tracked as KPI.
Claims Agent	Route complex claims to senior adjusters with full context	One-click routing with AI report, photos, annotations, notes.
Sr. Adjuster	Review AI + agent assessment for approval	Unified view. Full audit trail with timestamps.
Compliance	Audit any claim decision for QCB/SAMA review	Immutable log exportable per jurisdiction. RBAC enforced.

3. Key Features & Prioritization

RICE-prioritized. P0 = core AI-agent loop. P1 = accuracy + compliance. P2 = stakeholder extension.

P0: Must Have

- AI Damage Detection** - Vision model detects damage areas, classifies types, assigns severity.
- Cost Estimation** - Mapped to CCC ONE/Mitchell by vehicle spec and regional labor rates (Gulf luxury/SUV-weighted).
- Human-AI Review + Annotation** - Agents review, adjust, annotate damage regions (confirm/reject/add). Overrides enter training pipeline.
- Confidence-Based Routing** - Configurable thresholds for auto-approve vs. senior review. Government clients set stricter defaults.

P1: Should Have

- Multi-Photo Ensemble + Comparable Claims + RBAC** (agent, adjuster, auditor, admin tiers).

P2: Nice to Have

- Policyholder Self-Service. Arabic-First RTL UI. Najm/Claims Aggregator Integration** for Gulf platforms.

4. Success Metrics

KPI	Baseline	Target	Measurement
Assessment Time	~45 min	<5 min	Median upload-to-estimate
Estimate Accuracy	N/A	80% within 10%	AI vs. actual repair invoice
Agent Throughput	~12/day	~50/day	Claims per agent per shift
Auto-Approval Rate	0%	60% eligible	% without senior review
Model Drift	N/A	<2%/quarter	Accuracy delta between retrains
Override Rate	N/A	Declining	% modified by agents (learning signal)
Audit Compliance	N/A	100%	Every decision auditable per QCB/SAMA

5. AI Integration & Human-AI Interaction

Architecture

- Stage 1** - Fine-tuned YOLOv8 for damage localization. Training data weighted for Gulf vehicle mix (luxury sedans, SUVs).
- Stage 2** - Multi-modal LLM for contextual assessment: severity, repair feasibility, summary (English + Arabic).
- Cost Mapping** - Regional repair databases, Gulf labor rates, OEM parts availability.

Co-Pilot, Not Autopilot

AI Proposes, Human Disposes - Every assessment is a recommendation. Agents accept, modify, or override.

Interactive Annotation - Confirm, reject, add damage regions on images. Annotations enter Scale AI's training pipeline. The data flywheel.

Graduated Autonomy - Thresholds adapt on accuracy. Government clients enforce stricter review above configurable claim values.

Transparency - Highlighted regions, confidence breakdown, comparable claims, repair code references. No black box.

6. Data Strategy & Deployment

Training Data - Licensed datasets + client claims. Ongoing: agent annotations via Scale AI's pipeline. Gulf-specific: luxury vehicle and sand/heat damage patterns.

Quality - Inter-annotator agreement tracked. 5% random audit. **Class Imbalance** - Gulf over-indexes luxury (Land Cruiser, Patrol, Lexus). Active learning enriches long tail.

Data Sovereignty - On-premise or sovereign cloud within national boundaries. No cross-border transfer without QCB/SAMA authorization. Compliant with Qatar Data Privacy Law and Saudi PDPL.

7. Risks & Mitigations

Risk	Impact	Mitigation
AI confidently wrong on high-value claim	Liability, harm	Senior review >\$5K. Full audit trail for all AI-assisted decisions.
Adversarial / fraudulent submission	Financial loss	EXIF + geolocation validation. Fraud detection. Human review flag.
Model drift	Silent accuracy loss	Monthly retrain. <2%/quarter drift KPI. Automated alerting.
Gulf vehicle class bias	Luxury mis-estimation	Gulf-weighted training. Stratified eval by class. OEM parts pricing.
Cross-jurisdictional variance	QCB/SAMA failure	Per-jurisdiction configs. Regulatory export formats. RBAC data boundaries.

Governance: Immutable audit trail. RBAC (agent, adjuster, auditor, admin). Jurisdiction-specific export for QCB/SAMA. Bias monitoring. Human override always available. Qatar Data Privacy Law and Saudi PDPL compliant.